

Fixing India's Broken Grape Economy

A Scalable Model for Stable Farmer Income

Cluster-Based | Data Driven | Export-Optimized Model

High Yield $\neq$ High Income

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India's Grape Supply Chain is Profitable, Yet Broken



Key Facts

- Rs. 15 L/ha revenue potential
- 20% price crash in one season
- 15% yield loss due to climate shocks
- Majority farmers < 2 hectares (fragmentation)
- Maharashtra contributes 70%+ production and 63%+ exports



Problem Extension

- Supply chain disruptions (logistics, storage)
- Price volatility due to export shocks
- Trust deficit and delayed payments

“The issue is not production - it is system inefficiency.”

Multi - Layer Supply Chain Failures

Risk Impact Formula:

Price crash X Volume X Storage gap = Total Loss

Root Causes:

1. Supply Chain Inefficiency
 - Fragmentation
 - Logistics bottlenecks
 - Cold chain gaps
2. Market Dependency
 - Export reliance
 - Price crashes during shocks (covid evidence)
3. Coordination and Trust Gap
 - Payment delays
 - Lack of transparency



Top Risks (High Impact)

- Price crash (affect all farmers)
- Climate variability (reduces yield)
- Export barriers (affect premium markets)

“Pricing + storage inefficiency drives majority losses.”

Cluster Scoring Framework

Factor	Weight
Production Volume	30%
Export Readiness	25%
Infrastructure	20%
Climate Risk	15%
Market Access	10%

Cluster Identification:

- Tier 1 - Nashik (export)
- Tier 2 - Sangli, Pune
- Tier 3 - Karnataka, Tamil Nadu

Sourcing Mix:

- 60% Tier 1
- 25% Tier 2
- 15% Tier 3

Data-Driven Clustering-Based Sourcing Model

FPO Role (from research):

- Aggregation
- Quality control
- Market Linkage

Stakeholders:

- Farmers
- FPOs / FPCs
- Logistic players



Integrated Tech-Enabled Supply Chain System

01



Supply Layer

- FPO - based aggregation
- Pre - cooling & cold storage
- Efficient logistics

02

Data Intelligence Layer

- Inputs : 1. Price trends
2. Weather data
3. Export demand
- Outputs : 1. Harvest timing
2. Sell vs Store decision
3. Export allocation

03



Circular Economy Layer

- Grade A - Export
- Grade B - Domestic retail
- Grade C - Processing
(raisins, juice, wine,
compost / biomass)

04

Farmer Dashboard

- Price forecast
- Storage
recommendations
- Quality alerts



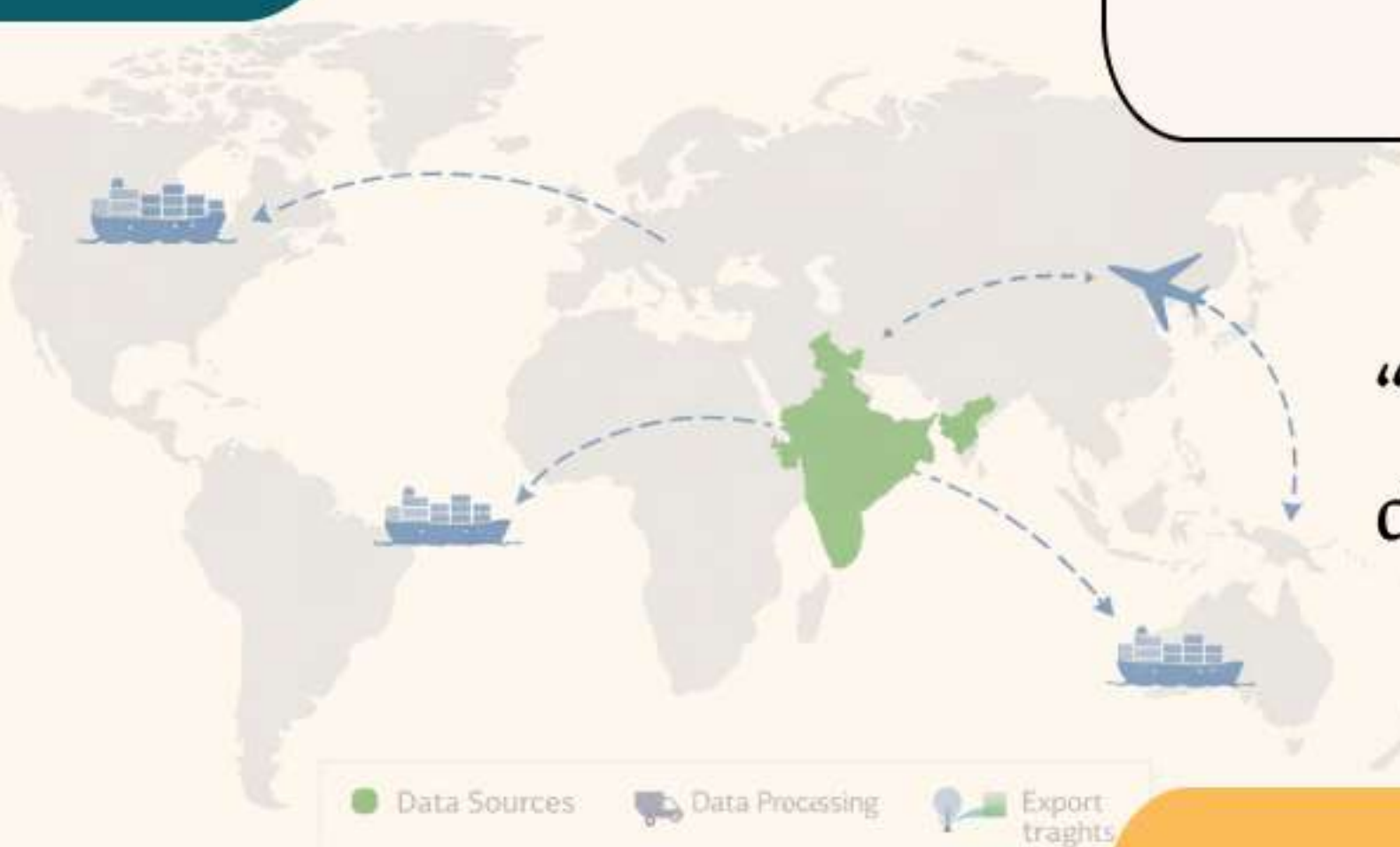


Before vs After:

Metric	Current	After
Farmer Income	Rs. 15L/ha	Rs. 18-20L/ha
Wastage	25-30%	10-15%
Price Volatility	High	40% (↓)

★ Value Improvement Breakdown

- Baseline Revenue : Rs. 15L/ha
- Current Losses : 1. wastage (25%) - ~Rs. 3.75L
2. price crash (20%) - ~ Rs. 2.1L
- Realized Income : ~Rs. 9-10L
- Post Solution Gains : 1. wastage ↓ by 10% - ~Rs. 2.25L
2. price optimization - ~Rs. 1L
3. export premium - ~Rs. 0.9L
- Net Income : ~Rs. 13-14L



Profitable, Stable & Sustainable System



Price Crash

Causes

- Oversupply
- No storage
- Export shocks

Solutions

- Internal - 1. demand forecasting
2. storage optimization
- External - 1. export diversification
2. forward contracts
- Digital - 1. real time price alerts
2. predictive analytics

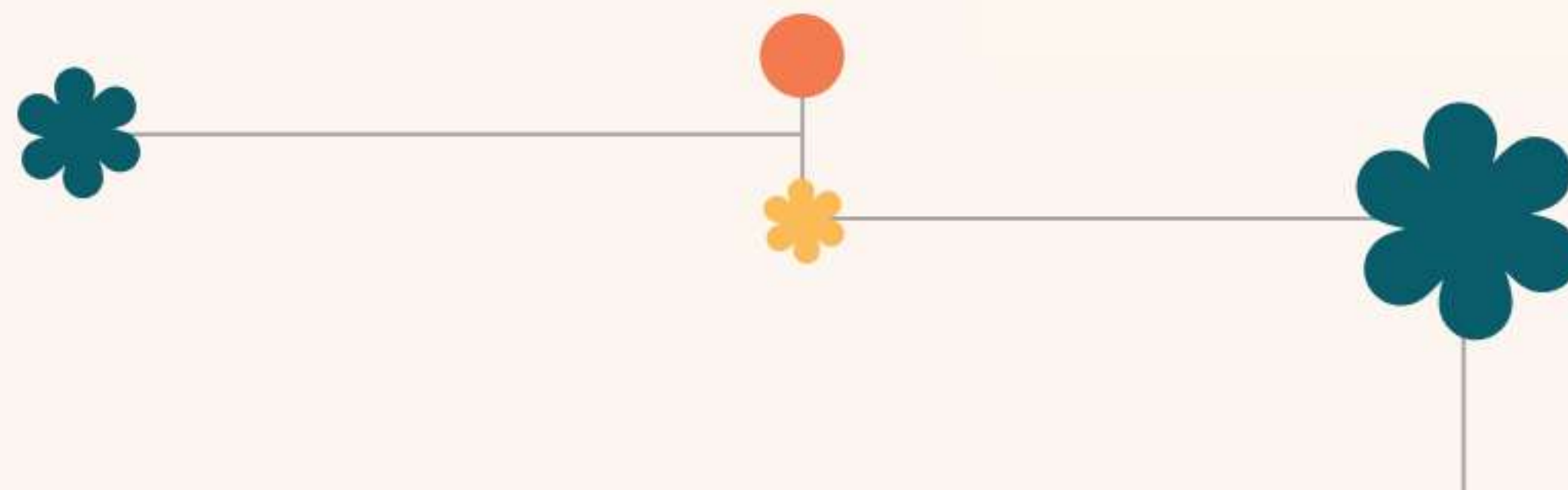
“From fragmented farming to a resilient, data-driven ecosystem.”

TBL Framework

- Environmental
- Efficient resource use
 - Reduced wastage

- Social
- FPO empowerment
 - Stable farmer income

- Economic
- Export premium
 - New revenue streams



Farmer

Farmer

- Pre-harvest contractor
- FPO / Aggregation
- Sorting and Grading
- **Decision 1 - Grade (Export / Domestic / Processing)**
- Pre-cooling (within 6 hrs)
- Cold storage
- **Decision 2 - Sell vs Store (data driven)**
- Transport
- Export / Domestic Market
- Retail → Consumer

Stage	Time	Cost	Value Add
Pre-cooling	4-6 hrs	Rs. 1-2 /kg	Shelf life ↑
Storage	7-30 days	Rs. 2-3 /kg	Price ↑
Logistics	10-15 days	Rs. 8-10 / kg	Premium ↑

1. Source : Tier 1 clusters
2. Variety : Thompson seedless (~79%)
3. Route : India to Europe (premium markets)

“Decision-making, not movement, drives value.”

Supporting Data, Risk & Strategy Mapping



Risk Mitigation

Environmental

- Climate forecasting
- Crop insurance

Labour

- Mechanization
- Skill training

Geographical

- Multi - region sourcing

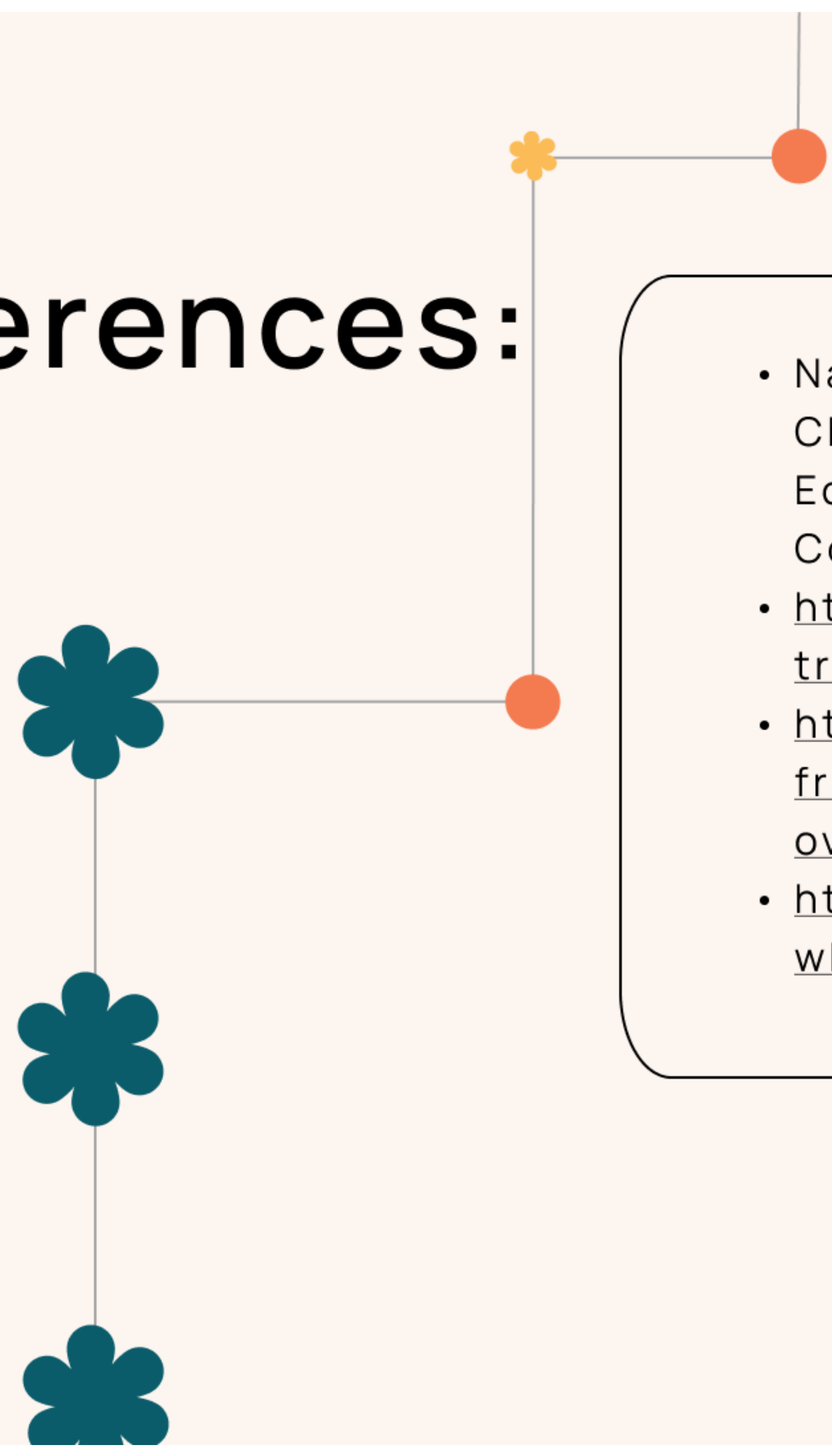
Economic

- Price hedging
- Market diversification

Key Data Points

- Avg Yield: 30 tons/hectare
- Avg Price: Rs. 50/kg → Rs. 15L/hectare revenue
- Export Volume: Rs. 3,460 Cr
- Price Drop: 20% (2023-24)
- Yield Loss: 15% due to climate

References:



- Navigating Challenges for Circular Grape Supply Chains: Blockchain, Business Ethics, and Socio-Economic Sustainability in Farmer Producer Companies of Maharashtra
- <https://tracextech.com/grape-supply-chain-traceability-with-tracex-for-exporters>
- <https://agriculture.institute/production-tech-fruit-crops/global-indian-grape-production-overview/>
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THANK YOU

“Fixing inefficiency today.
Securing farmer income tomorrow.”

